

Hayeon Jeong

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RESEARCH INTERESTS

Generative Models (T2I Diffusion), RL for Generation, Mechanistic Interpretability, AI Safety/Reliability.

EDUCATION

Yonsei University

Seoul, Korea

M.S. in the School of Integrated Technology

Mar. 2025 – Feb. 2027 (Expected)

- Advisor: Prof. Jong-Seok Lee (Multimedia Computing & Machine Learning Group)
- GPA: 4.16 / 4.30

University of California, Berkeley

Berkeley, CA, USA

Visiting Undergraduate Student

Jan. 2023 – May 2023

- Coursework: CS61A (Structure and Interpretation of Computer Programs), CS61B (Data Structures), Physics 77 (Computational Techniques in Physics).

Soongsil University

Seoul, Korea

B.S. in Electronic and Information Engineering (IT Convergence)

Mar. 2020 – Feb. 2025

- GPA: 4.11 / 4.50 (major 4.09 / 4.50); **Magna Cum Laude**
- Relevant coursework: Machine Learning, Deep Learning, Reinforcement Learning, Data Science, Probability & Random Variables, Linear Algebra, Algorithms, Computer Architecture.

RESEARCH EXPERIENCE

University of California, Los Angeles (UCLA)

Los Angeles, CA, USA

Visiting Researcher (Advisor: Prof. Cho-Jui Hsieh)

May 2026 – Oct. 2026 (Expected)

- *Flip-GRPO: Constraint-Level Counterfactual Pairs for Reinforcement Post-Training of Text-to-Image Models (on-going)*: Building a reinforcement-learning framework on top of a real-user prompt's simple, checkable requirements (e.g., is this object present, is the color correct, is the count correct, is the relation correct). Flips only the requirements the model currently fails to form counterfactual pairs, training the model to better satisfy each original requirement relative to its flipped counterfactual while preserving performance on the requirements it already satisfies.

Multimedia Computing & Machine Learning (MCML) Group, Yonsei University

Seoul, Korea

M.S. Researcher (Advisor: Prof. Jong-Seok Lee)

Mar. 2025 – Feb. 2027 (Expected)

- *Concept-level generative collapse in diffusion models (under review, first author)*: Identified and formalized the “dominant-vs-dominated” imbalance in multi-concept diffusion generation; built the DominanceBench benchmark and showed, via controlled fine-tuning and cross-attention head ablation, that low visual diversity of a concept's training data causally drives dominance.

AI Security Lab, Soongsil University

Seoul, Korea

Undergraduate Research Intern (Advisor: Prof. Daeseon Choi)

Jun. 2023 – Aug. 2024

- Built a thermal-RGB multimodal pipeline that localizes and mitigates person-detection failures using selective-search segmentation of thermal images; evaluated on a self-constructed physical-world dataset (AVSS 2025).
- Explored whether a generative model can serve as a backup device against attacks that encrypt user storage, by fine-tuning Stable Diffusion (DreamBooth) on user images and measuring how faithfully it reconstructs them; found that captioning the training images with an image-to-text model (BLIP-2) yields higher reconstruction fidelity than manually written descriptions.
- Proposed a privacy-assistant framework using small LLMs through systematic evaluation of their privacy-protection capability (Best Paper Award, KIISC Chungcheong Conference).

PUBLICATIONS

- **Hayeon Jeong** and Jong-Seok Lee, “[Dominating vs. Dominated: Concept-Level Generative Collapse in Diffusion Models](#),” *ACCV 2026* (under review); arXiv preprint, 2025.
- Sohee Park, **Hayeon Jeong**, and Daeseon Choi, “[Infrared Thermal-Guided Adversarial Patch Defense for Robust Visible Person Detectors](#),” *IEEE AVSS 2025*.

SELECTED PROJECTS

- **Border Conditions in Adversarial Patch Attacks.** Analyzed how patch border color, thickness, and interpolation method affect attack success rates against object detectors (Best Paper Award, 2nd Place, Undergraduate Thesis Competition, Soongsil University).
- **Phonetic-Aware Encoder Tuning for L2 Korean ASR.** Robust ASR framework mitigating phonetic disruption in non-native Korean speech via a LoRA-adapted Whisper encoder with auxiliary CTC-based phonetic supervision. ([project page](#))
- **Ctrl-F.** Generative model controlling ControlNet to resolve facial-distortion in human image generation.
- **Lightweight DenseNet.** EMNIST classifier with a DenseNet variant using $10\times$ fewer parameters than ResNet-50 and half the training time, and $2\times$ faster than DenseNet-121.

HONORS & AWARDS

- **Best Paper Award (KISA President’s Award)**, KIISC Chungcheong Conference, 2024.
- **Professor’s Scholarship** (25% tuition, merit-based), School of Electronic Engineering, Soongsil University, 2024.
- **Truth Award – Professor’s Scholarship** (50% tuition, merit-based), Soongsil University, 2024.
- Professor’s Scholarship (25% tuition, merit-based), School of Electronic Engineering, Soongsil University, 2021.

TEACHING & SERVICE

- Reviewer, *IEEE ICASSP 2026*.
- Teaching Assistant, *Advanced Engineering Mathematics*, Yonsei University, Spring 2025.

SKILLS

- **Languages:** Korean (native), English (TOEFL 4.5 (≈ 90 , old ver)).
- **Programming:** Python, Java, C/C++.
- **Frameworks:** PyTorch, Hugging Face (Diffusers/Transformers).
- **Tools:** Git, LaTeX.

OTHER EXPERIENCE

- Trainee, SK Hynix Semiconductor Curriculum 2.0, SUNI $\times$ SK hynix (Sep. 2023 – Dec. 2023); completed coursework on the semiconductor market, memory devices (DRAM, NAND), CIS, and future semiconductor trends.
- Data Annotator, AI Training Data Construction Project, Univa (Jul. 2022 – Sep. 2022); annotated mathematical-equation images to build supervised training data for an OCR recognition model.
- Member, Baird Volunteer Corps, Soongsil University (Oct. 2020 – Feb. 2022); contributed to community safety through a CCTV installation project with the local police station and youth education volunteer programs.
- Mentor, KT Mentoring Program (Oct. 2020 – Mar. 2022); developed a 6-month Math/English curriculum for mentees.